Interference Analysis of NGSO Constellation to GEO Satellite Communication System Based on Spatio–Temporal Slices

Di Yan, Yuanzhi He, and Huajun Fu

***Abstract*—The co-frequency interference (CFI) of the nongeostationary orbit (NGSO) constellation to the geostationary Earth orbit (GEO) satellite Internet of Things (IoT) system is an nonnegligible problem. In this article, based on the calculation method of the NGSO constellation satellite distribution state period, we propose a global characteristic method to analyze the CFI of the NGSO constellation to the GEO satellite IoT system by spatio-temporal slices. In order to analyze and evaluate the global characteristic and overall situation of interference, the temporal and spatial cut set of interference is established by fragmenting time and gridding space. The correctness of the proposed NGSO constellation satellite distribution state period is verified by taking the OneWeb constellation's interference to the SINOSAT-5 satellite communication system as the simulation scenario. Furthermore, the global interference characteristic of the OneWeb constellation to the SINOSAT-5 satellite communication system is analyzed with the method mentioned above in the range of 80 °E–140 °E and 0 °N–50 °N. The results show that the analysis method can reflect the overall situation of the CFI of the NGSO constellation to the GEO satellite IoT system.**

***Index Terms*—Co-frequency interference (CFI), interference analysis, nongeostationary orbit (NGSO) constellation, satellite communications, satellite Internet of Things (IoT).**

I. Introduction

IN RECENT years, the vigorously developing satellite communication technology is advancing from the single geostationary Earth orbit (GEO), medium Earth orbit (MEO) communication systems to multiorbit satellite communication systems that combine high and low orbit, and from the single satellite relay communication to the integrated satellite and Earth network communication. In particular, the rapid development of the micro-satellite manufacturing technology and the substantial reduction of satellite launch costs make nongeostationary orbit (NGSO) constellation communication network become an essential part of the future 6G wireless network architecture [1], [2], [3], [4], [5]. Nowadays, with the development of the Internet of Things (IoT), satellite communication network also provides a promising solution to support the IoT services. Many IoT systems based on the GEO satellite communication systems and NGSO satellite communication network have appeared successively [6], [7], [8], [9], [10], [11]. The frequency resources used by the emerging NGSO communication constellation systems are mainly concentrated in the Ku/Ka frequency band, which is also one of the frequency bands for the GEO satellites-based satellite-ground communication of the IoT [12], [13]. The deployment of the NGSO communication constellation systems is highly likely to cause harmful interference to the GEO satellite IoT systems [14], [15], [16]. Therefore, it is necessary to analyze the co-frequency interference (CFI) of the NGSO constellation to the GEO satellite IoT systems.

Since the 1970s, with the rapid development of the NGSO satellite communication system, scientists and engineers began to pay attention to the interference between different satellite communication systems, and some international scholars started to pay attention to the interference analysis of NGSO satellites to GEO satellite communication systems. Kobayashi et al. [17] began to research the interference between NGSO mobile satellite service (MSS) gateway station and GEO fixed satellite service (FSS) Earth station early. The inverse frequency band sharing method is proposed in the article, and the simulation results considering the dynamic movement of the NGSO gateway station antenna are given. However, the coordination process of the method is complicated, and the influence of the actual environment is not considered in the simulation. Park et al. [18] analyzed the influence of the number of satellites and the relative angle change on the bit error rate in the interference scene of the NGSO satellite communication system to the GEO satellite communication system. It analyzed the interference effect of the NGSO link on the GEO link from the perspective of bit error rate performance. However, the interference analysis model constructed in this literature only considered angular distance. Sharma et al. [19] studied the collinear interference scenario of the NGSO satellite communication system to the GEO satellite communication system, considered the spectrum coexistence of the GEO link and the NGSO link operating in the Ka-band, and proposed an adaptive power control technology to reduce the in-line interference. However, this method targets only a single or a few NGSO satellites and GEO satellites. Zhang et al. [20] introduced a specific method to reduce collinear interference by controlling the direction of the phased array antenna of tilted low-orbit satellites and analyzed the variation rule of the normal vector of phased array antenna direction to find the optimal direction. However, it is difficult

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to realize the real-time adjustment of phased array antenna direction on low-orbit satellites, and the technology is not mature. Wang et al. [21] analyzed the co-existing downlink interference between the LEO and GEO satellite communication systems. It evaluated the impact of the exclusive angle strategy on the interference. The proposed exclusive angle strategy model is relatively simple and only applies to a few NGSO satellites and GEO satellites. Jin et al. [22] established the mathematical model for analyzing interference among NGSO satellite communication systems. It proposed a link angle probabilistic analysis method for satellite constellation interference analysis and the probability calculation method of harmful interference among constellations and the availability index of constellations. Fortes et al. [23] analyzed the CFI among GEO satellite communication systems, designed the assessment scenarios of downlink and uplink interference caused by the global distribution of Earth stations and satellites, and proposed an evaluation method based on the extreme value of interference function. Simulation results proved the effectiveness and feasibility of the proposed method. However, none of the above literature analyzes the overall situation of the interference of the whole NGSO constellation to the GEO satellite communication system, and the previous research is only limited to the analysis of the instantaneous characteristics and partial characteristics of the interference, and lacks analysis from the perspective of the global characteristics of the interference constellation.

The analysis of the instantaneous and partial characteristics of the CFI can reflect the timely situation of the interference, which is conducive to the timely identification of the CFI level and is the basis for the subsequent work of harmful interference avoidance and mitigation. Meanwhile, the analysis of the global characteristics of the interference constellation can reflect the overall interference situation beyond the time limit, which is a new perspective for the objective analysis of the interference situation and can provide a basis for the constellation planning, ground terminal deployment, and other works.

Based on the above analysis, this article profoundly studies the analysis of CFI of the NGSO constellation on the downlink of GEO satellite IoT. Aiming at the problem that the traditional analysis methods only analyze the interference constellation's instantaneous and partial interference characteristics, the interference analysis scenario and mathematical analysis model are constructed. The calculation method of the NGSO constellation satellite distribution state period is given. On this basis, a global characteristic analysis method of CFI of NGSO constellation to GEO satellite IoT based on spatio-temporal slices is proposed. In this article, the simulation scenarios of interference caused by the OneWeb constellation to the SINOSAT-5 satellite communication system are constructed, and the correctness of the above-mentioned NGSO constellation satellite distribution state period is verified. Furthermore, the proposed method is used to analyze the global interference characteristics of the OneWeb constellation to the SINOSAT-5 satellite communication system in the range of 80 °E–140 °E and 0 °N–50 °N. According to the obtained overall interference intensity and outage probability, the layout scheme of the GEO IoT terminals is given.

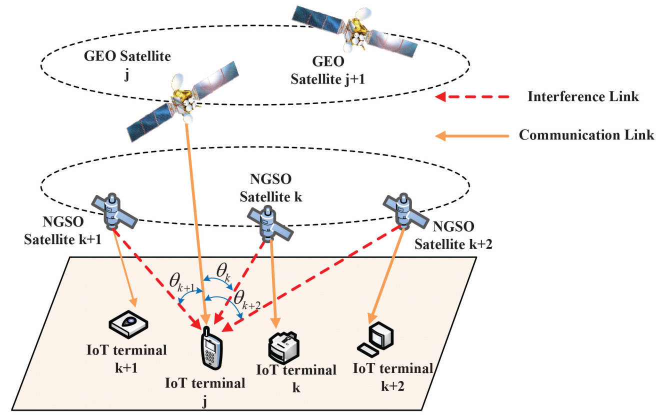


Fig. 1. Scene diagram of in-line interference of NGSO constellation to the downlinks of GEO satellite IoT systems.

II. System Model

A. Interference Scenario

The satellite IoT system based on the NGSO constellation uses a large amount of Ku/Ka frequency resources in satellite-ground communication, and its spectrum sharing with the GEO satellite IoT system has become a common phenomenon [24]. Therefore, in the case that NGSO satellites use the same communication frequency as GEO satellites, NGSO satellites' communication downlinks are collinear with the GEO satellite communication downlinks in a particular range when NGSO satellites run within the visible range of the GEO satellite IoT terminal, which causes in-line interference to the GEO communication downlinks. For the GEO satellite IoT system, IoT terminals tend to have larger antennas and higher power compared to NGSO satellite terminals. In this scenario, for the inbound (terminal to gateway via satellite) service of satellite IoT terminal, the interference signal power of the NGSO satellite terminal is smaller compared with the uplink signal power of the GEO satellite terminal. While for the outbound (gateway to terminal via satellite) service of the satellite IoT terminal, the interference signal power of the NGSO constellation is relatively larger. Therefore, this article only analyzes the in-line interference of the NGSO constellation to the downlink of the GEO satellite IoT system for the time being [25].

In the communication scenario shown in Fig. 1, the space segment includes GEO satellites and an NGSO constellation composed of several Walker constellations, and the ground segment includes a GEO satellite IoT terminal and several randomly distributed NGSO satellite IoT terminals. Assuming that the spindle of the GEO satellite IoT terminal's receiving antenna is always aimed at the GEO satellite, the spindles of the GEO and NGSO satellite's transmitting antenna are perpendicular to the ground, and the downlinks of the NGSO satellite IoT systems and GEO satellite IoT systems use the same frequency. In Fig. 1, the yellow solid lines represent the downlinks of the satellite-to-ground link of the GEO satellite IoT systems, and the red dotted lines represent the interference links caused by the NGSO satellites to the GEO satellite IoT terminals. The included angle between the interference and communication links is defined as θ. When θ is less than half of the interference discrimination angle θ_{th} of the GEO satellite IoT terminals, it is considered that the interference link has nonnegligible interference in the downlink of the GEO satellite IoT communication systems, and the corresponding NGSO satellites are called the major interference satellites to the GEO satellite IoT terminals.

B. Interference Discrimination Region

According to the interference discrimination angle θ_{th} of the receiving antenna, the interference discrimination region of the GEO satellite IoT terminal is defined as a conical region with the GEO satellite IoT terminal as its vertex, the spindle of the receiving antenna as the conical axis, and θ_{th} as the apex angle. The intersection region of this cone and the sphere surface (orbit shell of the constellation) is the interference discrimination region of the GEO satellite IoT terminal on each orbit shell of the NGSO constellation. The schematic diagram of Interference discrimination region in Geocentric rectangular coordinate system shown in Fig. 2.

The following is a mathematical expression of the interference discrimination region on the NGSO orbital spherical surface. Set the coordinates of the GEO satellite IoT terminal G as (L, B, H) in the geodetic coordinate system, where L represents the geodetic longitude of the satellite IoT terminal, B represents the geodetic latitude of the satellite IoT terminal, and H represents the geodetic height of the satellite IoT terminal. The geocentric rectangular coordinate system (X, Y, Z) is established, and the origin O is set at the mass center of the Earth. The x axis coincides with the intersection of the first meridian plane and the equatorial plane and is favorable to the East. The z axis coincides with the Earth's rotation axis and points to the north pole. The y axis is perpendicular to the xz plane to form a right-handed system [26].

Convert the coordinates of the GEO satellite IoT terminal in the geodetic coordinate system into coordinates in the geocentric cartesian coordinate system, and the conversion equations are as follows:

$$\begin{cases} x_0 = (R + H)\cos B \cos L \\ y_0 = (R + H)\cos B \sin L \\ z_0 = \left[R\left(1 - e^2\right) + H\right]\sin B \end{cases} \tag{1}$$

where R is the radius of the unitary circle, $R = (a/[\sqrt{1 - e^2 \sin^2 b}])$, $e^2 = ([a^2 - b^2]/a^2)$, a is the long half axis of the Earth's ellipsoid, and b is the short half axis of the Earth's ellipsoid.

The expression for a particular orbital shell of the NGSO constellation in the geocentric cartesian coordinate system is

$$x^2 + y^2 + z^2 = R_{\text{NGSO}}^2 \tag{2}$$

where R_{NGSO} is the radius of this orbit.

Let the coordinates of an NGSO satellite be $(x_{\text{NGSO}}, y_{\text{NGSO}}, z_{\text{NGSO}})$. So

$$x_{\text{NGSO}}{}^2 + y_{\text{NGSO}}{}^2 + z_{\text{NGSO}}{}^2 = R_{\text{NGSO}}{}^2. \tag{3}$$

Set the GEO satellite that corresponds to the GEO satellite IoT terminal as the point S in the geocentric rectangular coordinate system and its coordinates are $(x_{\text{GEO}}, y_{\text{GEO}}, 0)$, that

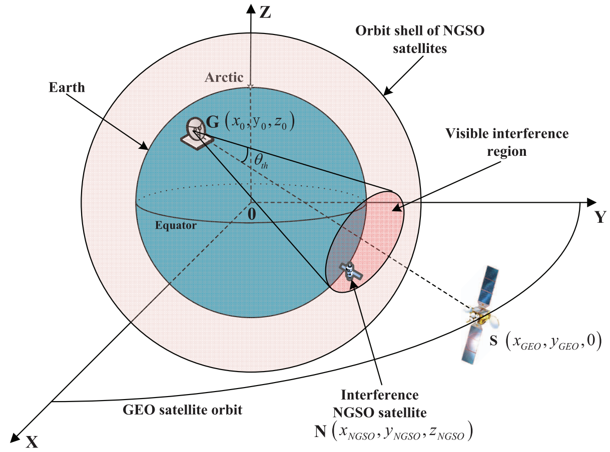


Fig. 2. Schematic diagram of Interference discrimination region in Geocentric rectangular coordinate system.

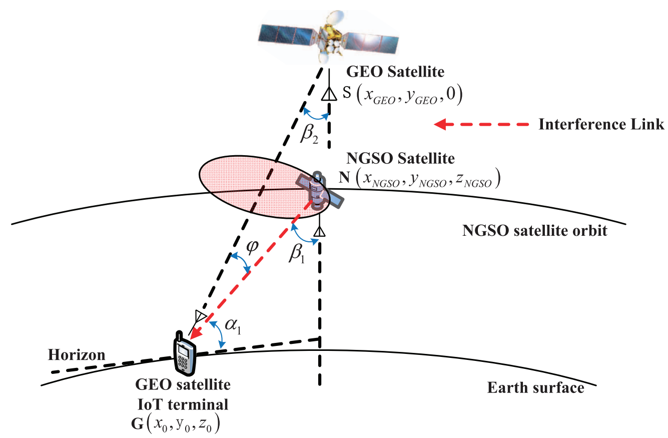


Fig. 3. Plane diagram of the interference link.

is, the GEO satellite is located on the equatorial plane. The expression of the line GS is

$$l_{GS} : \frac{x - x_{\text{GEO}}}{x_{\text{GEO}} - x_0} = \frac{y - y_{\text{GEO}}}{y_{\text{GEO}} - y_0} = \frac{z}{-z_0}. \tag{4}$$

Meanwhile, suppose that point (x_t, y_t, z) is a point on the ray GS, then the conical surface with point G as the vertex, GS as the axis line, and θ_{th} as the apex angle can be expressed as (5), shown at the bottom of the next page.

By combining (5) with (2), the interference discrimination region of the GEO satellite IoT terminal on each orbit shell of the NGSO constellation can be obtained. The shape and size of the interference discrimination region vary with the latitude and longitude of GEO satellite IoT terminals and the pointing of receiving antennas.

C. Downlink Model

As shown in Fig. 3, for NGSO satellites that have moved to the interference discrimination region of the GEO satellite IoT terminal, assuming that the transmission power of the NGSO satellites is I_s, then the interference signal power I_r of the NGSO satellites received by the GEO satellite IoT terminal can be expressed as follows:

$$I_r = I_s + G_{\text{NGSO}}(\beta_1) + G_r(\varphi) - L_{\text{NGSO}}. \tag{6}$$

In (6), the units of I_s and I_r are dBW; $G_{\text{NGSO}}(\beta_1)$ dBi refers to the transmission gain of the NGSO satellite antenna; $G_r(\varphi)$ dBi refers to the receiving gain of the GEO satellite IoT terminal antenna; L_{NGSO} dB represents the signal transmission loss of the communication link.

Suppose the radiation pattern for the NGSO satellite antenna with a circular or elliptical beam is impossible to measure. In that case, we can refer to the satellite antenna pattern provided by ITU-R S.1528 [27] for interference analysis. The reference pattern $G_{\text{NGSO}}(\beta_1)$ for an NGSO satellite antenna having the antenna diameter to wavelength ratio ($D/\lambda \leq 35$) is given by (7) [28], [29]

$$G_{\text{NGSO}}(\beta_1) = \begin{cases} G_{1,\max} - 3(\beta_1/\phi_1)^2, & \phi_1 < \beta_1 < Y_s \\ G_{1,\max} + L_s - 25\log(\beta_1/Y_s), & Y_s < \beta_1 < Z_s \\ L_f, & Z_s < \beta_1 < 180^\circ. \end{cases} \tag{7}$$

For LEO satellites in the NGSO constellation, $L_s = -6.75$, $Y_s = 1.5\phi_1$

$$G_{\text{NGSO}}(\beta_1) = 20\log(D/\lambda) + 5.65 - 25\log(\beta_1/\phi_1), \quad 1.5\phi_1 < \beta_1 < Z_s \tag{8}$$

where β_1 deg refers to the off-axis angle of the NGSO satellite in the direction of the GEO satellite IoT terminal, and $\beta_1 = 90^\circ - \alpha_1 - \delta_1$, α_1 deg refers to the elevation angle of the NGSO satellite with respect to the GEO satellite IoT terminal, δ_1 deg refers to the geocentric angle between the NGSO satellite and the GEO satellite IoT terminal, $Y_s = \phi_1(-L_s/3)^{1/2}$, $Z_s = Y_s \times 10^{0.04}(G_{1,\max} + L_s)$, $G_{1,\max}$ dBi refers to the maximum gain of the main flap of NGSO satellite antenna, D meter is the antenna diameter, λ meter is the signal wavelength, ϕ_1 deg is one-half the 3-dB beamwidth of the NGSO satellite antenna, L_s dBi is near-in side-lobe level relative to the peak gain required by the system design and L_f dBi is 0 far-out side-lobe level.

For the antenna pattern of the GEO satellite IoT terminal, we can refer to ITU-RS.465 [30] for analysis. The calculation equations for the receiving gain of the GEO satellite IoT terminal antenna are as follows:

$$G_r(\varphi) = \begin{cases} 32 - 25\log\varphi, & \varphi_{\min} \leq \varphi < 48^\circ \\ -10, & 48^\circ \leq \varphi \leq 180^\circ. \end{cases} \tag{9}$$

When the antenna aperture diameter to wavelength ratio is less than 50 ($D_G/\lambda \leq 50$), $\varphi_{\min} = \max(2, 114(D_G/\lambda)^{-1.09})$ deg. In (9), φ deg refers to the off-axis angle of the GEO satellite IoT terminal in the direction of the NGSO satellite, which can be obtained from (10), shown at the bottom of the page.

The transmission loss of interference signal usually includes free space loss and propagation loss caused by atmospheric phenomena, such as clouds, fog, and rain. If the most severe interference is considered, it is necessary to assume that the attenuation of the interference signal in the transmission process is the smallest, so only the free space loss is considered as the transmission process [31]. Its calculation equation is

$$L_{\text{NGSO}} = 20\log(4\pi d/\lambda) = 32.4 + 20\log f + 20\log d \tag{11}$$

where f is the frequency of signal, in MHz; d is the link length, in km, and

$$d = \left|\overrightarrow{SG}\right| = \sqrt{(x_0 - x_s)^2 + (y_0 - y_s)^2 + z_0^2}. \tag{12}$$

Substituting (8), (9), and (11) into (6) gives

$$\begin{aligned} I_r &= I_s + G_{\text{NGSO}}(\beta_1) + G_r(\varphi) - L_{\text{NGSO}} \\ &= \begin{cases} I_s + 20\log\left(\frac{D}{d}\right) - 25\log\left(\frac{\varphi\beta_1}{\phi}\right) + 5.25, & \varphi_{\min} \leq \varphi < 48^\circ \\ I_s + 20\log\left(\frac{D}{d}\right) - 25\log\left(\frac{\beta_1}{\phi}\right) - 36.75, & 48^\circ \leq \varphi \leq 180^\circ. \end{cases} \end{aligned} \tag{13}$$

According to (13), the received interference signal power of an interfering satellite in the NGSO constellation in the interference discrimination region that interferes with the GEO satellite IoT terminal at a particular time can be obtained.

In a similar process, assuming that the GEO satellite transmission power is P_s, the GEO satellite useful signal power P_r received by the GEO satellite IoT terminal can be expressed as follows:

$$P_r = P_s + G_{\text{GEO}}(\beta_2) + G_{r,\max} - L_{\text{GEO}}. \tag{14}$$

In (14), the units of P_s and P_r are dBW; $G_{\text{GEO}}(\beta_2)$ dBi refers to the transmission gain of the GEO satellite antenna; $G_{r,\max}$ dBi refers to the maximum gain of the main flap of the GEO satellite IoT terminal antenna; L_{GEO} dB refers to the transmission loss of the useful signal.

The antenna pattern $G_{\text{GEO}}(\beta_2)$ for the GEO satellite referred to ITU-R.S.672 [32] can be expressed as follows:

$$G_{\text{GEO}}(\beta_2) = \begin{cases} G_{2,\max} - 3(\beta_2/\phi_2)^2, & \phi_2 < \beta_2 \leq a_2\phi_2 \\ G_{2,\max} + L_n, & a_2\phi_2 < \beta_2 \leq b_2\phi_2 \\ G_{2,\max} + L_n + 20- \\ 25\log(\beta_2/\phi_2), & b_2\phi_2 < \beta_2 < \gamma \\ 0, & \gamma \leq \beta_2 < 90^\circ \\ 3, & 90^\circ \leq \beta_2 < 180^\circ \end{cases} \tag{15}$$

$$S_{\text{cone}} : \begin{cases} \frac{x - x_{\text{GEO}}}{x_{\text{GEO}} - x_0} = \frac{y - y_{\text{GEO}}}{y_{\text{GEO}} - y_0} = \frac{z}{-z_0} \\ \left[\tan\left(\frac{\theta_{th}}{2}\right)\sqrt{(x_t - x_0)^2 + (y_t - y_0)^2 + (z_t - z_0)^2}\right]^2 = (x - x_t)^2 + (y - y_t)^2 + (z - z_t)^2 \end{cases} \tag{5}$$

$$\begin{aligned} \varphi &= \arccos\left(\frac{\overrightarrow{GS} \cdot \overrightarrow{GN}}{\left|\overrightarrow{GS}\right|\left|\overrightarrow{GN}\right|}\right) \\ &= \arccos\left(\frac{(x_0 - x_{\text{GEO}})(x_0 - x_{\text{NGSO}}) + (y_0 - y_{\text{GEO}})(y_0 - y_{\text{NGSO}}) + z_0(z_0 - z_{\text{NGSO}})}{\sqrt{(x_0 - x_{\text{GEO}})^2 + (y_0 - x_{\text{GEO}})^2 + z_0^2}\sqrt{(x_0 - x_{\text{NGSO}})^2 + (y_0 - y_{\text{NGSO}})^2 + (z_0 - z_{\text{NGSO}})^2}}\right) \end{aligned} \tag{10}$$

where β_2 deg refers to the off-axis angle of the GEO satellite in the direction of the GEO satellite IoT terminal, and $\beta_2 = 90° - \alpha_2 - \delta_2$, α_2 deg refers to the elevation angle of the GEO satellite with respect to the GEO satellite IoT terminal, δ_2 deg refers to the geocentric angle between the GEO satellite and the GEO satellite IoT terminal; $G_{2,\max}$ dBi refers to the maximum gain of the main flap of the GEO satellite antenna; ϕ_2 deg is one-half the 3-dB beamwidth of the GEO satellite antenna, L_n is the near-in-side-lobe level in dB relative to the peak gain required by the system design; γ is the value of β_2 when $G_{\mathrm{GEO}}(\beta_2) = G_{2,\max} + L_n + 20 - 25\log(\beta_2/\phi_2)$ equals to 0 dB. For $L_n = -20$, the values of a_2 and b_2 are 2.58 and 6.32, respectively.

According to (15), the received power of useful signal sent by the GEO satellite to the GEO satellite IoT terminal at a particular time can be obtained.

D. Downlink Outage Probability of GEO Satellite IoT System

Outage probability is one of the essential indicators to measure the quality of communication links. The downlink outage constraint of the GEO satellite IoT system is

$$P_{\text{outage}} = \Pr\{R_0 \le C_0\} \le \eta \tag{16}$$

where R_0 is the data rate of the GEO satellite IoT system, C_0 is the minimum rate required for the GEO satellite IoT terminal, and η is the downlink outage threshold of the GEO satellite IoT system. If $P_{\text{outage}} > \eta$, the downlink of the GEO satellite IoT system is interrupted. According to the Shannon capacity, the GEO satellite IoT system's achievable rate can be written as follows [33]:

$$R_0 = W\log_2\left(1 + \frac{P_r}{I_{ag} + N_{\text{noise}}}\right) \tag{17}$$

where P_r represents the received power of the GEO satellite, I_{ag} represents the aggregated interference power of NGSO satellites, W is communication bandwidth, and denotes the noise that can be calculated by $N_{\text{noise}} = kTW$, where k is a Boltzmann constant and T is the temperature of the GEO satellite IoT terminal. Therefore, by substituting (17) into (16), the downlink outage probability of the GEO satellite IoT system can be written as follows:

$$P_{\text{outage}} = \Pr\{R_0 \le C_0\} = \Pr\left\{I_{\text{ag}} \ge \frac{P_r}{2^{C_0/W} - 1} - N_{\text{noise}}\right\}. \tag{18}$$

III. Interference Analysis Method Based on Spatio-Temporal Slices

Based on Section II, a global characteristic analysis method of CFI of NGSO constellation to GEO satellite IoT based on spatio-temporal slices is proposed, aiming to analyze the overall interference of the NGSO constellation on large time scales.

A. NGSO Constellation Satellite Distribution State Period

According to the constellation configuration and orbit characteristics of the NGSO constellation, the NGSO constellation satellite distribution state period is defined as the unit time scale of interference analysis.

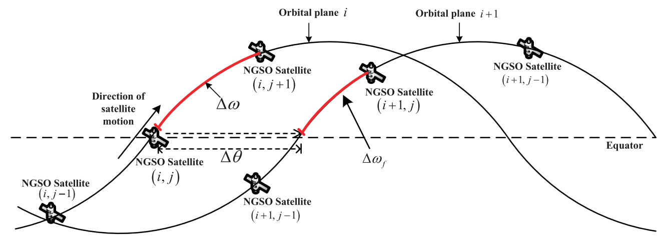


Fig. 4. Schematic diagram of the phase difference between corresponding satellites of adjacent orbits in the Walker constellation.

Most of the NGSO constellations consist of one or more Walker constellations with evenly distributed satellites in each orbit, and all satellites within the same Walker constellation can be considered to have the same communication functions.

When an NGSO constellation consists of only one Walker constellation, the Walker code ($N_W/P_W/F_W : h_W, i_W$) is usually used to describe the configuration of the Walker constellation, where N_W represents the total number of satellites in the Walker constellation, P_W represents the number of orbital planes in the Walker constellation, h_W represents the orbital height, i_W represents the orbital inclination, and F_W is called phase factor which can determine the phase differences between the corresponding satellites in adjacent orbits in the Walker constellation.

Fig. 4 illustrates the phase differences between the corresponding satellites in adjacent orbits in the Walker constellation. In the figure, the two curves, respectively, represent the tracks of the subsatellite point of two adjacent orbital planes i and $i+1$. $\Delta\theta$ represents the longitude differences between the ascending node of two adjacent orbital planes. The bold red part of the track of the subsatellite point of orbital planes $i+1$ represents the phase differences between the corresponding satellites in adjacent orbits $\Delta\omega_f$, whose measurement method takes the intersection of the latitude circle of the satellite on the orbital plane i and the orbital plane $i+1$ as the starting point, and the first satellite on the orbital plane $i+1$ along the direction of satellite motion as the corresponding phase of the endpoint. Given $\Delta\omega_f = 2\pi(F_W/N_W)$, then, after $T_{\min} = (\Delta\omega_f/2\pi)\cdot T_{\text{orbital}}$ time, the satellite in the latter orbital plane can move to the distribution state of the satellite at the beginning of the previous orbital plane, where T_{orbital} represents an orbital period. At the same time, after $T_{\min}$, the difference in longitude between the ascending node of the orbital plane at the equator is $\Delta\theta_{\min} = (T_{\min}/T_E)\cdot 360$, where T_E represents a sidereal day. Let the least common multiple of $\Delta\theta$ and $\Delta\theta_{\min}$ be $\Delta\theta_T$, i.e., $\Delta\theta_T = \lfloor\Delta\theta, \Delta\theta_{\min}\rfloor$. $\Delta\theta_T$ represents the longitude value change of the ascending node of the orbital plane when the satellite distribution state of the entire Walker constellation changes back to the initial state for a certain point on the Earth. In other words, when the ascending node of the orbital plane of the entire Walker constellation moves $\Delta\theta_T$ longitude on the equator, the satellite distribution state of the entire Walker constellation is completely consistent with that before the movement. We call this period the satellite distribution state period of the Walker constellation, which is recorded as T_{Walker}

$$T_{\text{Walker}} = \frac{\Delta\theta_T T_E}{360} = \frac{T_E}{360}\left\lfloor \frac{360}{P_W}, \frac{360 F_W T_W}{N_W T_E} \right\rfloor \tag{19}$$

where $\lfloor , \rfloor$ represents the least common multiple of the two elements, T_E represents the Earth's rotation period, and T represents the orbital period of this Walker constellation. For a fixed Earth station on the ground, the satellite distribution status of the entire Walker constellation, the number of visible satellites in the Walker constellation, and the changes of visible satellite trajectories all change with time according to the T_{Walker}.

When the NGSO constellation is composed of several different Walker constellations, let the NGSO constellation be composed of δ different Walker constellations, and note that the satellite distribution state period of each Walker constellation is T^i_{Walker}, where $i = 1, 2, \ldots, \delta$, then the whole NGSO constellation satellite distribution state period T_{NGSO} is

$$T_{\text{NGSO}} = \left\lfloor T^1_{\text{Walker}}, T^2_{\text{Walker}}, \ldots, T^\delta_{\text{Walker}} \right\rfloor \tag{20}$$

where $\lfloor, \ldots, \rfloor$ represents the least common multiple of each element.

According to the above analysis, for a fixed satellite IoT terminal on the ground, the satellite distribution state of the whole NGSO constellation, the number of visible satellites in the NGSO constellation, and the change of visible satellites' motion trajectory all change with time according to T_{NGSO} cycle. The satellite distribution state period of the NGSO constellation is taken as the time unit to analyze the interference of the NGSO constellation in order to analyze the overall interference characteristics of the NGSO constellation to the GEO satellite communication system on a large time scale.

B. Interference Analysis Based on Spatio-Temporal Slices

For a given NGSO constellation, the NGSO constellation satellite distribution state period T_{NGSO} can be calculated according to (19) or (20), and the interference of the NGSO constellation to a GEO satellite IoT terminal can be analyzed within the period $[0, T_{\text{NGSO}}]$.

Set an analysis step t and traverse all NGSO satellites at every time t starting from the moment 0. Each traversal is stopped until a reference satellite is found. If there is no reference satellite, the next traversal will be carried out. Assume that the reference satellite at a particular time is $K_{i,j}$, which means the jth NGSO satellite in the ith orbital plane, and the longitude and latitude coordinates of the subsatellite point are $(L^0_{\text{NGSO}}, B^0_{\text{NGSO}})$. According to (1), we can get

$$\begin{cases} x^0_{i,j} = R_{\text{NGSO}} \cos B^0_{\text{NGSO}} \cos L^0_{\text{NGSO}} \\ y^0_{i,j} = R_{\text{NGSO}} \cos B^0_{\text{NGSO}} \sin L^0_{\text{NGSO}} \\ z^0_{i,j} = R_{\text{NGSO}} \sin B^0_{\text{NGSO}} \end{cases} \tag{21}$$

where $(x^0_{i,j}, y^0_{i,j}, z^0_{i,j})$ represents the coordinates of the reference satellite in the geocentric rectangular coordinate system. If $(x^0_{i,j}, y^0_{i,j}, z^0_{i,j})$ is substituted into (5), the coordinates of the reference satellite should satisfy (22), shown at the bottom of the page.

With the reference satellite $K_{i,j}$ as the center, we examine whether its neighboring satellites are within the interference discrimination range from near to far. The pseudo code for the major interference satellites discrimination algorithm is shown in Algorithm 1. Suppose a satellite to be investigated is numbered as $K_{i+\Delta i, j+\Delta j}$, where Δi represents the number of changes in the orbital plane and Δj represents the number of changes in the orbital plane. The longitude and latitude coordinates $(L'_{\text{NGSO}}, B'_{\text{NGSO}})$ of the points under the satellite $K_{i+\Delta i, j+\Delta j}$ are

$$\begin{cases} L'_{\text{NGSO}} = L^0_{\text{NGSO}} + (2\pi\Delta i)/P_W \\ B'_{\text{NGSO}} = B^0_{\text{NGSO}} + 2\pi \sin(i_{\text{NGSO}})\left[(F_W/N_W)\Delta i + (P_W/N_W)\Delta j\right]. \end{cases} \tag{23}$$

Then, determine whether $K_{i+\Delta i, j+\Delta j}$ is within the visible interference region according to the methods provided by (21) and (22).

According to the above analysis method, the time of a satellite distribution state period is sliced, and the time slices sets are constructed by class. For a particular GEO satellite IoT terminal, the maximum number of major interference satellites is N_m, and the duration of each interference satellite within the interference discrimination region is expressed by $T_m = [T_{m0} \quad T_{m1} \quad \cdots \quad T_{mN_m}]$, where T_{mn}, $n = 0, 1, \ldots, N_m$ represents the time segment when the number of major interference satellites for the terminal m is n, which is expressed as follows:

$$T_{mn} = \left[(t_1, t_2)_{\text{mn}} \quad (t_3, t_4)_{\text{mn}} \quad \cdots \quad (t_{2k-1}, t_{2k})_{\text{mn}} \right] \quad k = 1, 2, \ldots, K_{mn} \tag{24}$$

where K_{mn} represents the total number of time fragments when the number of major interference satellites for the terminal m is n, and $(t_{2k-1}, t_{2k})_{mn}$ represents the period from moment t_{2k-1} to moment t_{2k}.

In an NGSO constellation satellite distribution state period, the probability when the number of major interference satellites of the terminal m is n can be obtained as follows:

$$P_m(n) = \frac{\sum_{k=1}^{K_{mn}} (t_{k+1} - t_k)}{T_{\text{NGSO}}}. \tag{25}$$

In combination with (13) and (24), for a certain GEO satellite IoT terminal m, the mean value of the aggregated interference power can be expressed as follows:

$$U_m = \left[u_{m1} \quad u_{m2} \quad \cdots \quad u_{mN_m} \right] \tag{26}$$

where $u_n (n = 0, 1, \ldots, N_m)$ represents the mean value of the aggregated interference power received by the terminal m when the number of major interference satellites is n, and

$$\left[\tan\left(\frac{\theta}{2}\right)\sqrt{(x_t - x_0)^2 + (y_t - y_0)^2 + (z_t - z_0)^2}\right]^2 \geq \left(x^0_{i,j} - x_t\right)^2 + \left(y^0_{i,j} - y_t\right)^2 + \left(z^0_{i,j} - z_t\right)^2 \tag{22}$$

Algorithm 1 Pseudo Code for the Major Interference Satellites Discrimination Algorithm

Input:
Walker code, $(N_W/P_W/F_W : h_{\text{NGSO}}, i_{\text{NGSO}})$;
NGSO satellite mark number, (i, j);
Interference discrimination region, S_{in};
Output:
The number of major interference satellites, δ;
function SINGLEORBIT(i)
  $m \leftarrow 0$
  for $\Delta j = 1 \rightarrow \lfloor \frac{N}{2P} \rfloor$ **do**
    if $K_{i,j+\Delta j} \notin S_{\text{in}}$ **and** $K_{i,j-\Delta j} \notin S_{\text{in}}$ **then**
      break
    else if $K_{i,j+\Delta j} \in S_{\text{in}}$ **then**
      $m \leftarrow m + 1$
    else if $K_{i,j-\Delta j} \in S_{\text{in}}$ **then**
      $m \leftarrow m + 1$
    end if
  end for
  return m
end function
function MULTIORBIT($walkercode, i, j, S_{\text{in}}$)
  $\delta \leftarrow 1 + singleorbit(i)$
  for $\Delta i = 1 \rightarrow \lfloor \frac{P}{2} \rfloor$ **do**
    if $K_{i+\Delta \text{i},j} \notin S_{\text{in}}$**and**$K_{i-\Delta \text{i},j} \notin S_{\text{in}}$ **then**
      break
    else if $K_{i+\Delta \text{i},j} \in S_{\text{in}}$ **then**
      $\delta \leftarrow 1 + singleorbit(i + \Delta i)$
    else if $K_{i-\Delta \text{i},j} \in S_{\text{in}}$ **then**
      $\delta \leftarrow 1 + singleorbit(i - \Delta i)$
    end if
  end for
  return δ
end function

its calculation equation is (27), shown at the bottom of the page.

Equation (27) divides an NGSO constellation satellite distribution state period according to the number of major interference satellites and describes the interference of the whole NGSO constellation to the GEO satellite IoT terminal from the time dimension.

Combine (27) with (25), the overall interference intensity of the GEO satellite IoT terminal m disturbed by the signals of the whole NGSO constellation in an NGSO constellation satellite distribution state period can be obtained as (28), shown at the bottom of the page.

According to (18), the probability of downlink outage of terminal m due to NGSO constellation's interference is (29), shown at the bottom of the page.

Equation (29) describes the overall interference intensity of the whole NGSO constellation to the GEO satellite IoT terminal m. For different terminals in the beam coverage area of the GEO satellite, $P_{\text{outage}}(m)$ is different due to different geographical locations. The coverage area of the GEO satellite is divided into M grids according to certain granularity, and the interference situation of this grid is represented by the interference situation of the satellite IoT terminal in the center of the grid area. The location set of each GEO satellite IoT terminal is expressed as $\mathbf{B} = [B_1\ B_2\ \cdots\ B_M]^T$, the maximum quantity set of major interference satellites is expressed as $\mathbf{N} = [N_1\ N_2\ \cdots\ N_M]^T$, interference duration set is expressed as $\mathbf{T} = [T_1\ T_2\ \cdots\ T_M]^T$ and the outage probability set is expressed as $\mathbf{P} = [P_{\text{outage}}(1)\ \ P_{\text{outage}}(2)\ \ \cdots\ \ P_{\text{outage}}(M)]^T$. Then, the disturbance $\mathbf{I}$ of the whole coverage area can be expressed as follows:

$$\mathbf{I} = \begin{bmatrix} \mathbf{B} & \mathbf{N} & \mathbf{T} & \mathbf{P} \end{bmatrix} = \begin{bmatrix} B_1 & N_1 & T_1 & P_{\text{outage}}(1) \\ B_2 & N_2 & T_2 & P_{\text{outage}}(2) \\ \vdots & \vdots & \vdots & \vdots \\ B_M & N_M & T_M & P_{\text{outage}}(M) \end{bmatrix}. \tag{30}$$

IV. Simulation Verification and Analysis

First, verify the correctness of the proposed NGSO satellite distribution state period.

STK 11.2 simulation software is used in the simulation to build the simulation scenario of the OneWeb LEO satellite communication system's interference to the SINOSAT-5 GEO satellite communication system. SINOSAT-5 satellite is located at 110.5 °E in geostationary orbit, and its satellite network data ID in the ITU satellite network database is 106 520 145 [34]. OneWeb LEO satellite communication system is a Walker constellation with an orbital height of

$$u_{mn} = 10\log \frac{\sum_{k=1}^{K_{mn}} \int_{t_k}^{t_{k+1}} \left(10^{\left(\frac{I_r{}^1(t)}{10}\right)} + 10^{\left(\frac{I_r{}^2(t)}{10}\right)} + \cdots + 10^{\left(\frac{I_r{}^n(t)}{10}\right)}\right) d(t)}{\sum_{k=1}^{K_{mn}} (t_{k+1} - t_k)} \tag{27}$$

$$I_r(m) = 10\log\left(10^{\left(\frac{u_{m1}}{10}\right)} P_m(1) + 10^{\left(\frac{u_{m2}}{10}\right)} P_m(2) + \cdots + 10^{\left(\frac{u_{mN_m}}{10}\right)} P_m(N_m)\right) \tag{28}$$

$$P_{\text{outage}}(m) = \Pr\{R_0 \leq C_0\} = \sum_{n=1}^{N_m} \Pr\left\{u_{mn} \geq \frac{P_r}{2^{C_0/W} - 1} - N_{\text{noise}}\right\} \tag{29}$$

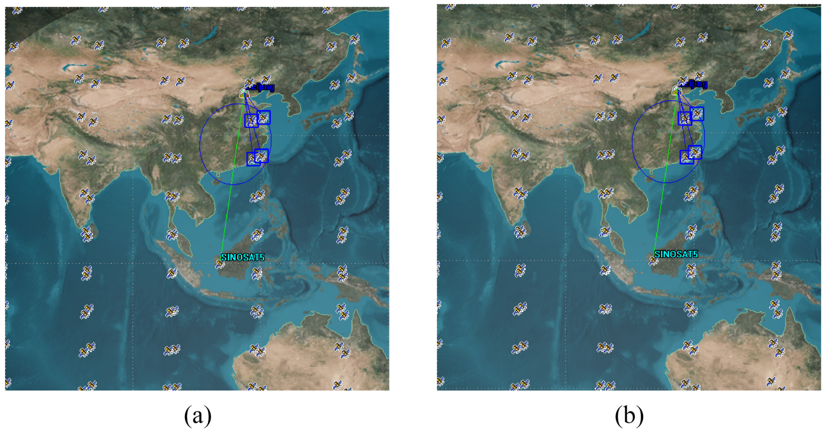


Fig. 5. 2-D map of NGSO satellite distribution when the simulation time is (a) 0 s and (b) 4787 s.

1200 km and an orbital inclination of 87.9°. It has 18 orbital planes and 40 satellites in each orbital plane. The satellite network data ID of the OneWeb system is 113 520 120 [35]. In the simulation, a GEO satellite IoT terminal is set in Beijing (40.12 °N, 116.23 °E) with an altitude of 38 m. The interference discrimination angle of the receiving antenna of the GEO satellite IoT terminal is assumed to be 35°. All satellite transmitting antenna spindles of the OneWeb system point to the Earth center, causing interference to the downlink of the SINOSAT-5 satellite communication system.

In the above simulation scenario, the NGSO constellation is composed of a walker constellation. According to (19), the calculated NGSO constellation satellite distribution state period is 4787 s, 1 h 19 min 47 s. The satellite distribution when the simulation time is 0 s and the simulation time is 4787 s is shown in Fig. 5. The distribution of the satellite when the simulation time is 4787 s is the same as when the simulation time is 0 s. Then, within the simulation time of 30 000 s, the link distances of all major interference satellites for the GEO satellite IoT terminal in Beijing and their changes with time are obtained, as shown in Fig. 6. The horizontal axis in the figure is time, in seconds, and the vertical axis is the sum of the link distances of all major interference satellites at any time, in meters. It can be seen from the figure that the satellite distribution of the NGSO constellation has obvious periodicity. The time difference between the six marked points in the figure is 4760, 4917, 4760, 4750, and 4760 s, respectively, with an average of 4789 s, which is only 2S different from the calculated NGSO constellation satellite distribution state period of 4787 s, which verifies that the calculated NGSO constellation satellite distribution state period is accurate.

Based on the above conclusions, the proposed interference analysis method of the NGSO constellation to the GEO satellite IoT system based on spatio-temporal slices is used to analyze the interference of the simulation scene. The parameter settings in the simulation are shown in Table I.

Fig. 7 shows the aggregate interference average power of the OneWeb constellation interference signals received by the terminal in Beijing at each number of major interference satellites during one NGSO constellation satellite distribution state period, and Fig. 8 shows the occurrence probability of each number of major interference satellites.

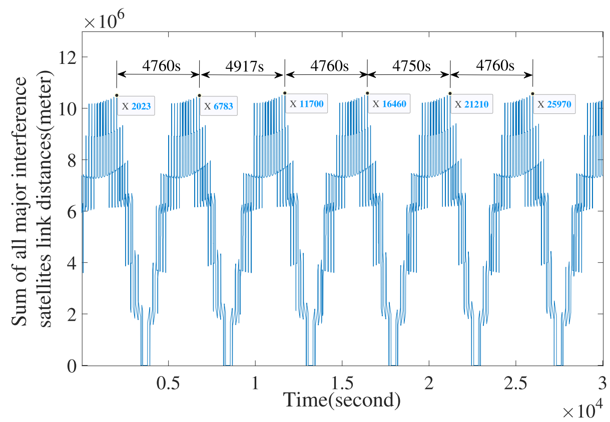


Fig. 6. Link distances and distances of all major interference satellites of GEO Earth station at any time within 30 000 s.

TABLE I
SIMULATION PARAMETERS [4], [14], [22], [36], [37]

Parameter	Parameter setting
Orbit prediction model	TwoBody
Link type	Downlink
OneWeb satellite antenna peak gain /dBi	25.9
OneWeb satellite antenna HPBW /(°)	7.9
OneWeb satellite transmitting power /dBW	8.5
OneWeb satellite antenna model	ITU-R S.1528
SINOSAT-5 satellite antenna peak gain /dBi	37
SINOSAT-5 satellite transmit power /dBW	16.3
SINOSAT-5 satellite antenna model	ITU-R S.672
SINOSAT-5 IoT terminal antenna peak gain /dBi	32.9
SINOSAT-5 IoT terminal antenna HPBW /(°)	3.5
SINOSAT-5 IoT terminal interference discrimination angle /(°)	30
SINOSAT-5 IoT terminal antenna model	ITU-R.S.465
Communication frequency /GHz	11.51
Communication bandwidth/MHz	125
the minimum rate required for terminals/Kbps	64
the temperature of the GEO satellite IoT terminal/K	200

According to analysis based on Figs. 7 and 8, the OneWeb constellation has at most five visible interference satellites to the terminal in Beijing, and the occurrence probability of five visible interference satellites is the highest. The aggregated interference average power when there are five visible interference satellites is the highest. The overall interference

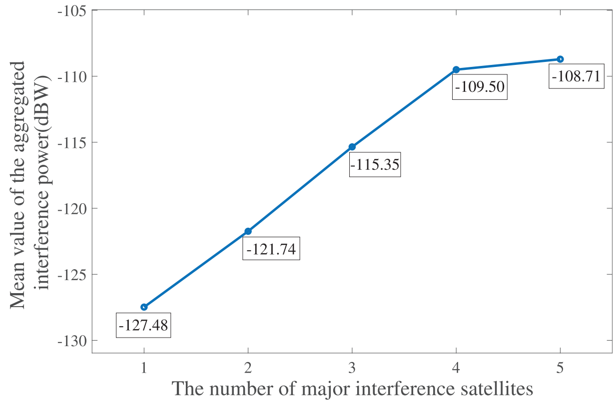


Fig. 7. Mean value of aggregated interference power under the different number of major interference satellites.

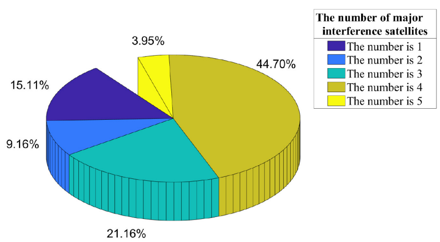


Fig. 8. Probability of occurrence of the different number of major interference satellites.

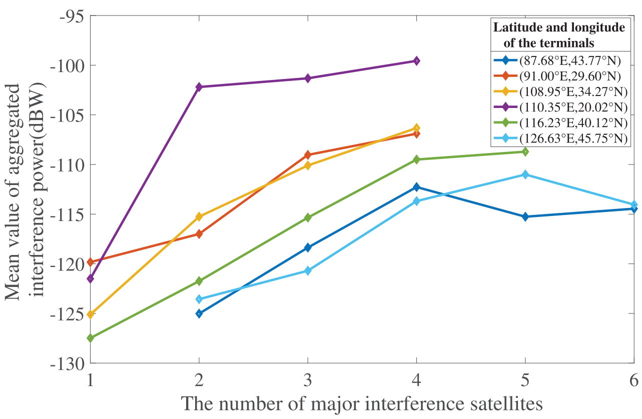


Fig. 9. IoT terminals in six different cities are interfered with the OneWeb constellation.

intensity of the constellation to the terminal in Beijing is calculated as −112.40 dBW by (28), and the outage probability is calculated as 0 by (29). That is, no communication outage occurs in this simulation setting.

Figs. 9 and 10 show the situation that IoT terminals located in six cities within the range of 80 °E–140 °E and 0 °N–50 °N are interfered with OneWeb constellation signals. It can be analyzed that the maximum number of visible interference satellites is smaller for the GEO satellite IoT terminal which is closer to the longitude of GEO satellite orbit, and the minimum number of visible interference satellites is larger for the GEO satellite IoT terminal which is further away from the longitude of GEO satellite orbit. This is because the interference discrimination range of the GEO satellite IoT terminal closer to the longitude of the GEO satellite orbit is smaller, and the

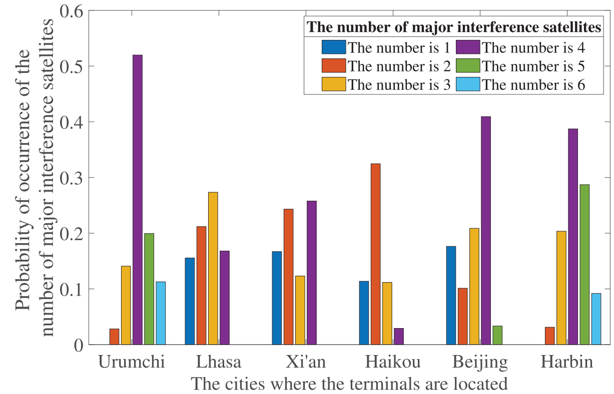
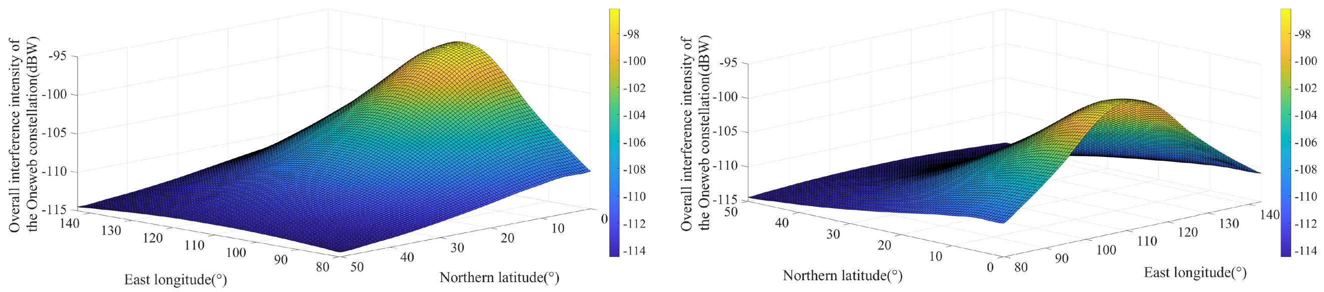


Fig. 10. Probability of occurrence of the number of major interference satellites of IoT terminals in six different cities.

interference discrimination range is significantly more influenced by the longitude of the terminal than by the dimension. It can be seen that the proposed method for analyzing the global characteristics of the CFI of the NGSO constellation to the GEO Satellite IoT system based on the space-time cut set can obtain the overall interference of the NGSO constellation to the GEO satellite IoT terminals at different locations beyond the time limit.

Figs. 11 and 12 show the overall interference intensity of the OneWeb constellation to IoT terminals within the range of 80 °E–140 °E and 0 °N–50 °N. It can be seen that within the ground coverage of the GEO satellite, the overall interference intensity increases with decreasing latitude. At the same latitude, the closer the IoT terminal is to the longitude of the GEO satellite orbit (110.5 °E), the greater the interference intensity will be. The interference intensity of the IoT terminal near 110.5 °E at low latitudes increases significantly, and the interference intensity of the whole area reaches the maximum value of –96.22 dBw at (110.5 °E, 0 °N). This shows that the influence of interference link distance on interference intensity is stronger than the influence of interference discrimination range. It can also be observed from Figs. 11 and 12 that the overall interference intensity is symmetric about the 110.5 °E meridian axis, which is because that the interference discrimination regions on the NGSO orbital spherical surface of the two GEO IoT terminals that are symmetric about the 110.5-°E meridian are also symmetric, and visible interference satellites of two GEO IoT terminals at a certain moment are different, but the overall interference situation in one NGSO constellation satellite distribution state period is the same. This also verifies the scientific nature of the proposed NGSO constellation satellite distribution state period and better reflects the overall characteristics of the interference analysis.

Figs. 13 and 14 give the probability that the downlink of IoT terminals in GEO satellite coverage area 90 °E–130 °E, 0 °N–30 °N is interrupted by interference from the NGSO constellation. It can be seen that no communication interruption occurs in the area above 14 °N due to the interference of the NGSO constellation, and the probability of communication interruption is higher at lower latitudes close to the longitude line of 110.5 °E. The outage probability reaches the maximum probability of 41.44% at (110.5 °E,



(a) (b)

Fig. 11. Overall interference intensity of the OneWeb constellation at different positions. (a) 3-D display map (main view is from high latitude to low latitude). (b) 3-D display map (main view from low latitude to high latitude).

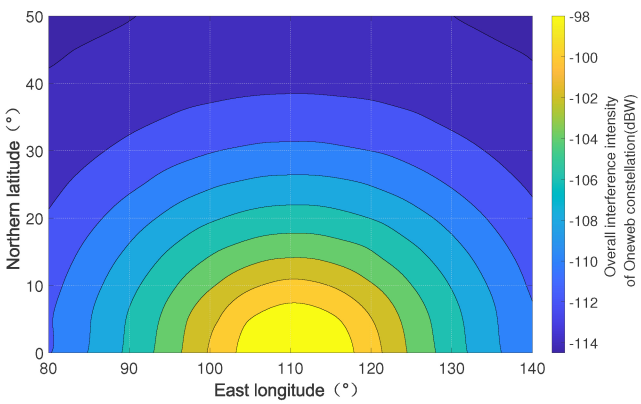


Fig. 12. Contour map of overall interference intensity of the OneWeb constellation at different positions.

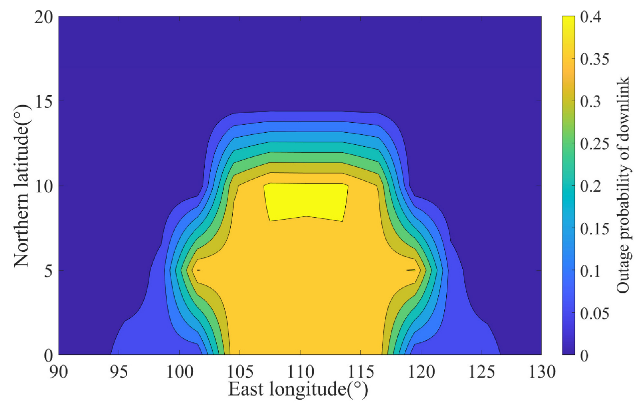


Fig. 14. Contour map of outage probability.

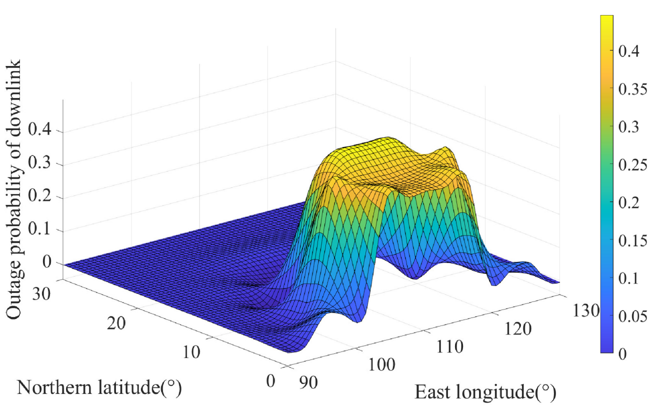


Fig. 13. Outage probability of downlink.

10 °N). Similarly, it is observed that the interference situation embodied by the outage probability is symmetric about the 110.5-°E meridian axis.

According to the above analysis results, it can be concluded that in this communication scenario, the deployment location of the GEO IoT terminals is selected in the area with 0 outage probability to avoid harmful interference from the NGSO constellation to the maximum extent. In particular, the accuracy of interference analysis can be adjusted by defining different interference discrimination angles of the GEO satellite IoT terminals in different communication scenarios. The smaller the interference discrimination angle, the higher the accuracy of the interference analysis. Adjusting interference discrimination angle and parameters, such as terminal demand rate will change the obtained outage probability and eventually affect the analysis results.

V. Conclusion

This article deeply studies the problem of CFI analysis of the NGSO constellation on the downlink of the GEO Satellite IoT system. Aiming at the problem that the traditional analysis methods only analyze the instantaneous characteristics of interference and the local interference characteristics of the interference constellation, the interference analysis scenario, and mathematical analysis model are constructed. The calculation method of the distribution state period of the NGSO constellation satellites is given. On this basis, a global characteristic analysis method of CFI between the NGSO constellation and GEO Satellite IoT system based on spatio-temporal slices is proposed. The simulation scenarios of interference caused by the OneWeb constellation to the SINOSAT-5 satellite communication system are constructed, respectively to verify the correctness of the proposed NGSO constellation satellite distribution state period. The proposed interference global characteristic analysis method is also used to analyze the overall interference characteristics of the OneWeb constellation to the SINOSAT-5 satellite communication system, and the overall interference conditions of target IoT terminals at different locations are analyzed. The overall interference intensity of the OneWeb constellation within the ground coverage of SINOSAT-5 and the probability of GEO IoT terminals interrupted by the interference of the NGSO constellation satellites are obtained.

The results can be used as a reference for analyzing the CFI of the NGSO constellation to the GEO satellite communication system, formulating avoidance measures, and providing support for developing the GEO IoT terminal layout scheme.

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